

TRUSTNET AI: AI-POWERED TRUST & THREAT INTELLIGENCE PLATFORM

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Abstract—Digital communication environments face an unprecedented surge in multi-channel cyber threats, including sophisticated phishing links, social engineering texts, malicious QR code redirects (quishing), and fraudulent financial identifiers. Conventional threat detection solutions analyze these attack vectors in isolated silos using opaque, binary machine learning classifiers that fail to surface underlying risk rationale or contextual evidence. This paper presents TrustNet AI, a novel unified threat intelligence platform designed to ingest and evaluate multi-modal digital signals—specifically URLs, natural language messages, QR code payloads, UPI payment handles, and telephone credentials. The proposed platform employs a hybrid analysis engine integrating machine learning classifiers, natural language processing, and graph network topology metrics (PageRank and degree centrality) to calculate a calibrated composite risk score. To resolve the black-box opacity inherent in traditional AI scoring, TrustNet AI incorporates a Retrieval-Augmented Generation (RAG) framework that anchors Large Language Model reasoning in authenticated threat advisories, producing human-auditable explanations alongside quantitative risk scores. Implemented as a high-throughput React frontend and FastAPI backend system, TrustNet AI delivers transparent, defensible, and actionable threat intelligence.

Index Terms—TrustNet AI, Threat Intelligence, Multi-Modal Analysis, Explainable AI, Risk Assessment, Retrieval-Augmented Generation (RAG).

I. INTRODUCTION

The rapid proliferation of digital communication channels, mobile financial services, and web-based financial payment platforms has fundamentally altered the landscape of personal and enterprise communications. While digital connectivity enables seamless global transaction flows, it has simultaneously expanded the cyber attack surface. Adversaries increasingly orchestrate multi-faceted threat campaigns targeting vulnerable end-users and organizational perimeters. Modern digital threats encompass a broad spectrum of malicious artifacts, including deceptive Uniform Resource Locator (URL) phishing schemes, coercive natural language text messages designed for social engineering, malicious Quick Response (QR) code redirects (commonly termed quishing), fraudulent Unified Payments Interface (UPI) virtual payment addresses, and spoofed telecommunication credentials.

A major systemic shortcoming of current cybersecurity solutions is their strict reliance on single-domain, isolated signal analysis. Modern cybercrime operations rarely present as standalone, isolated anomalies. Instead, threat actors organize coordinated cross-channel campaigns. For instance, an

adversary might initiate contact by dispatching an urgent SMS message containing a QR code. When scanned, the QR code redirects the victim to a credential-harvesting phishing website while facilitating illicit money transfers via a spoofed UPI handle. Traditional security tooling—such as standalone URL scanners, isolated spam SMS filters, or static bank black-lists—evaluates each artifact independently. Because these point solutions lack cross-domain correlation mechanisms, sophisticated evasion techniques frequently pass undetected through perimeter defenses.

Furthermore, standard machine learning classifiers used in commercial threat detection generally output binary classification labels (e.g., "Legitimate" vs. "Phishing") or uncalibrated numeric probabilities between 0.0 and 1.0. For security operations center (SOC) analysts, incident responders, and end-users, receiving an isolated numeric risk score without contextual justification is fundamentally inadequate. As highlighted in recent security usability literature, the core research challenge in modern digital threat response is not merely detection accuracy, but establishing why a given digital artifact poses risk ("Every verdict ships with the evidence behind it — not a black box score"). Without transparent feature attributions, interpretable evidence, and human-auditable narratives, analysts cannot validate automated alerts, conduct root-cause investigations, or defend security decisions against regulatory oversight.

To directly address these operational and research limitations, this paper introduces TrustNet AI, a comprehensive, AI-powered trust and threat intelligence platform. TrustNet AI establishes a unified intelligence layer capable of concurrently ingesting and processing five distinct digital signal modalities. The platform integrates supervised machine learning classifiers, natural language semantic extraction models, and Neo4j graph network centrality analytics (specifically PageRank and node degree centrality) to formulate calibrated, multi-signal risk assessments. Furthermore, TrustNet AI addresses the model interpretability challenge by integrating a Retrieval-Augmented Generation (RAG) architecture. By coupling a ChromaDB vector database with Large Language Models (LLMs), TrustNet AI produces plain-English, evidence-anchored threat explanations alongside quantitative risk scores.

The key scientific and engineering contributions of this work are summarized as follows:

- 1) *Unified Multi-Modal Architecture*: We propose a cohesive

intelligence platform capable of ingesting and correlating five heterogeneous digital signals (URLs, text messages, QR codes, UPI handles, and phone credentials) within a single unified processing pipeline.

2) *Hybrid Analysis & Graph Intelligence*: We combine supervised machine learning ensembles, natural language processing models, and graph network centrality algorithms (PageRank and degree centrality) to identify hidden structural relationships across threat actors and digital assets.

3) *Calibrated Risk Scoring & Feature Attribution*: We introduce a multi-signal risk fusion formulation that calculates calibrated threat probabilities paired with quantitative SHAP (SHapley Additive exPlanations) feature weight attributions.

4) *RAG-Grounded Explainable AI (XAI)*: We design a Retrieval-Augmented Generation workflow that anchors LLM textual explanations in verified cybersecurity threat advisories, preventing AI hallucinations in security-critical environments.

5) *Operational Production Web Interface*: We implement and deploy a high-performance web interface built on React, TypeScript, Tailwind CSS, and FastAPI microservices, providing an interactive operational dashboard for real-time threat investigation.

II. LITERATURE SURVEY

A. Existing Approaches

Digital threat detection has been widely investigated across isolated domain boundaries. In the domain of URL phishing detection, early approaches relied on static domain blacklists, WHOIS registration lookups, and manual heuristic rules. Sahingoz et al. [1] demonstrated that machine learning algorithms utilizing lexical URL patterns, host properties, and domain age achieve strong performance in identifying phishing sites. However, pure URL classifiers remain blind to social engineering context embedded in accompanying text messages or multi-channel communications.

Natural Language Processing (NLP) has made substantial progress in detecting scam and phishing text messages. Verma et al. [6] surveyed NLP techniques applied to social engineering attacks, demonstrating that linguistic urgency, authority manipulation, and financial solicitations serve as primary indicators of fraudulent intent. Transformer-based architectures, such as BERT [8], have enhanced semantic intent recognition. Nevertheless, text classifiers alone cannot verify whether referenced payment handles or destination URLs carry legitimate infrastructure reputations.

QR code security research (quishing) focuses on decoding visual matrix codes and evaluating embedded payloads. Kadam et al. [7] highlighted how adversaries exploit the visual opacity of matrix QR codes to hide malicious redirects. Current quishing mitigation relies primarily on post-decoding URL analysis, treating the QR code as a neutral transport layer rather than correlating it with sender identity or network history.

In graph network security, Akoglu et al. [4] and Wang et al. [5] demonstrated that graph-based anomaly detection algorithms—specifically node degree centrality, PageRank, and

relational graph neural networks—effectively uncover illicit financial rings and coordinated fraud syndicates. By modeling accounts, devices, and communication endpoints as graph nodes, structural relationships reveal anomalies that single-entity classifiers miss.

Model interpretability in security has gained prominence following the work of Lundberg and Lee [2] on SHAP feature weights, which quantify individual feature contributions to model outputs. In parallel, Lewis et al. [3] introduced Retrieval-Augmented Generation (RAG) to combine parametric memory from pre-trained language models with non-parametric vector database retrieval. While RAG has seen success in general enterprise search, its application to grounded cyber threat explanation remains an emerging research area.

B. Research Gap

Despite significant advancements in single-domain security tools, an explicit research gap persists in the literature: existing security solutions remain fragmented. Specialized models excel within narrow domains but operate independently without cross-domain signal fusion. Furthermore, traditional machine learning security models present a trade-off between predictive performance and explainability—complex ensemble architectures achieve high predictive power but function as black boxes, while simple decision trees offer interpretability at the expense of accuracy.

TrustNet AI directly addresses this research gap by establishing a unified multi-modal threat intelligence layer. By coupling supervised machine learning classifiers and graph centrality metrics with SHAP attributions and RAG-driven LLM explanations, TrustNet AI replaces black-box scoring with auditable, evidence-backed security verdicts.

III. PROPOSED SYSTEM ARCHITECTURE

A. Overall Architecture

The structural layout of TrustNet AI is designed as a scalable, modular microservice architecture. As illustrated in Fig. 1, the platform separates client interaction, API orchestration, analytical processing, machine learning inference, risk fusion, explainable reasoning, and dashboard delivery into decoupled operational tiers.



Fig. 1. Overall architecture of the proposed TrustNet AI platform.

The interaction flow begins at the *React Frontend*, which provides a high-performance web console for single-input or

TABLE I
COMPARATIVE SUMMARY OF EXISTING THREAT DETECTION LITERATURE VS. TRUSTNET AI

Author / Reference	Methodology / Approach	Target Domain	Primary Limitation
Sahingoz et al. (2019) [1]	Lexical & NLP feature extraction + ML classifiers	Phishing URL detection	Single-modal focus; cannot analyze message or payment context.
Lundberg & Lee (2017) [2]	SHAP additive feature attribution framework	General XAI interpretation	Surfaces numerical weights but lacks semantic natural language reasoning.
Lewis et al. (2020) [3]	Retrieval-Augmented Generation (Vector DB + LLM)	Knowledge-intensive NLP	General-purpose QA focus; not tailored to structured multi-signal threat evaluation.
Akoglu et al. (2015) [4]	Graph mining, PageRank, degree centrality algorithms	Entity anomaly detection	High computational complexity; lacks natural language explainability.
Wang et al. (2021) [5]	Graph Neural Networks (GNN) for threat intelligence	Multi-source threat fusion	Requires extensive static graph data; limited real-time web deployment capability.
Verma et al. (2022) [6]	Transformer NLP semantic text analysis	Social engineering SMS	Fails to evaluate external infrastructure (URLs, UPI handles, QR codes).
Kadam et al. (2023) [7]	Decoder-based matrix parsing & URL extraction	QR code security	Isolated analysis; lacks cross-modal signal fusion and graph verification.
Chen & Guestrin (2016) [9]	Gradient boosted decision tree ensemble (XGBoost)	Tabular classification	High accuracy but acts as a black box without explainability layers.
Ribeiro et al. (2016) [10]	Local Interpretable Model-agnostic Explanations (LIME)	Local classifier interpretation	Unstable sampling variance; lacks retrieval grounding in threat advisories.
Vaswani et al. (2017) [11]	Transformer self-attention architecture	Sequence Modeling	High resource requirement; black-box attention weights lack direct grounding.
Kouliaridis et al. (2020) [12]	Mobile fraud and telecom malware detection	Mobile Security	Focused on host OS telemetry rather than cross-modal threat signal fusion.
TrustNet AI (Proposed)	Multi-modal ML + Graph Centrality + SHAP + RAG	Unified Threat Intelligence	Addressed: Fuses 5 modalities with grounded XAI explanations.

batch signal submission. The request payload is transmitted via asynchronous REST APIs to the *FastAPI Backend* orchestration layer. FastAPI handles client authentication, input routing, request validation, and task queuing.

Once ingested, payloads are dispatched to the specialized *Analysis Engine*, comprising sub-modules for URL structural processing, Natural Language Processing (NLP), QR matrix payload extraction, and Graph Network analysis (Neo4j). Extracted feature representations are evaluated by the *ML / AI Engine* for model inference. The resulting sub-scores are aggregated within the *Risk & Trust Engine*, which executes signal fusion algorithms to calculate a calibrated composite risk score. Concurrently, the *Explainable AI / RAG Engine* retrieves relevant threat advisories and synthesizes plain-English risk narratives, returning complete evidence packages to the *Dashboard* for user presentation.

B. Multi-Modal Analysis

TrustNet AI implements dedicated analysis handlers across five distinct digital modalities, establishing a unified threat processing engine:

1) *URL Analysis Module*: Inspects link structure, domain age, WHOIS records, SSL/TLS certificate validity, subdomain depth, entropy scores, IP address redirects, and lexical similarity to top target brands (homograph/typosquatting detection).

2) *Message NLP Module*: Analyzes textual input for urgency indicators, coercive language patterns, credential requests,

authority impersonation, and structural matches against known scam phrasing templates.

3) *QR Code Module*: Decodes matrix visual data, parses embedded payload strings, identifies hidden URL redirects or payment protocols, and inspects visual code structural anomalies.

4) *UPI Payment Module*: Evaluates financial payment handles against known fraud registry databases, analyzing beneficiary handle structure, domain routing, and transaction velocity anomalies.

5) *Phone Credential Module*: Assesses telecommunication metadata, country code alignment, carrier reputation, reported spam activity, and voice/SMS scam patterns.

C. Threat Analysis Pipeline

The operational processing pipeline of TrustNet AI follows a rigorous eight-stage sequential workflow designed to narrow uncertainty until a defensible threat verdict is rendered, as depicted in Fig. 2.

The pipeline operates as follows: 1) Stage 01 — Input Acquisition: Accepts digital artifact inputs via the web console or external REST API endpoints. 2) Stage 02 — Input Validation: Sanitizes raw input data, strips malicious control characters, performs regex format verification, and normalizes canonical strings. 3) Stage 03 — Feature Extraction: Extracts structural, lexical, linguistic, and network features tailored to the identified modality. 4) Stage 04 — ML / NLP / Graph



Fig. 2. Threat analysis pipeline of TrustNet AI.

Analysis: Executes parallel inference across supervised classifiers (XGBoost), NLP intent transformers, and Neo4j graph centrality engines (PageRank and degree centrality). 5) Stage 05 — Signal Fusion: Integrates individual model confidence values and graph topology risk weights using an ensemble fusion formulation. 6) Stage 06 — Risk Score Calculation: Computes the final calibrated numeric risk percentage (0% to 100%) and categorizes the threat level (Low, Medium, High, Critical). 7) Stage 07 — Grounded Explanation: Triggers the RAG workflow to retrieve matching security advisories and synthesize a natural language explanation backed by SHAP feature weights. 8) Stage 08 — Final Result Delivery: Renders the complete threat intelligence card, evidence trace, and actionable recommendations on the user dashboard.

IV. IMPLEMENTATION AND METHODOLOGY

A. Input Acquisition & API Orchestration

Input acquisition is handled via asynchronous RESTful endpoints implemented in Python FastAPI. The ingestion layer supports single-artifact scans (e.g., submitting an isolated URL or SMS text) and batch telemetric feeds. Input request headers are authenticated using JSON Web Tokens (JWT) and API key pairs. Upon receiving a payload, FastAPI instantiates an asynchronous worker thread that dispatches request objects to parallel processing queues, enforcing sub-second API response SLAs.

B. Input Validation and Preprocessing

Prior to analysis, raw payloads undergo strict input validation and preprocessing. URLs are canonicalized by parsing schemes, decoding UTF-8 percent-encoded strings, resolving shortened redirects (e.g., bit.ly, tinyurl), and extracting root domain components via Public Suffix List parsing. Text messages undergo tokenization, lowercasing, stop-word removal, and Lemmatization. QR code images are processed using OpenCV and ZBar matrix decoders to extract raw byte payloads.

C. Feature Extraction Architecture

Feature extraction maps raw inputs into numerical vector spaces across four primary feature domains:

1) *Lexical & Structural Features*: URL length, character entropy, digit-to-letter ratio, path depth, query string parameters, subdomain count, and IP-based host presence.

2) *Linguistic & Semantic Features*: TF-IDF n-gram vectors, sentiment polarity, urgency keyword frequency vectors, and

transformer embedding vectors capturing social engineering intent.

3) *Graph Topology Features*: Node degree centrality (number of direct entity linkages), PageRank score (relative importance within the transaction network), and shortest-path distance to known high-risk seed nodes in Neo4j.

4) *Domain & Infrastructure Features*: Domain age in days (via WHOIS lookup), SSL certificate validity, HTTPS protocol enforcement, and DNS A/MX record consistency.

D. Threat Analysis Engine & Sub-Models

The threat analysis engine executes parallel model inference across specialized classifiers. Supervised classification is performed using an XGBoost gradient boosting ensemble trained on structured lexical and domain features. Natural language intent classification relies on fine-tuned transformer models evaluating semantic embeddings. Graph topology analysis executes Cypher queries against a Neo4j graph database to calculate PageRank and degree centrality metrics across linked entities.

E. Risk and Trust Scoring Formulation

To aggregate heterogeneous risk signals into a single, defensible metric, TrustNet AI employs a calibrated composite scoring formulation. The composite risk score $R_{composite}$ is defined as a weighted linear combination of individual modality scores adjusted by graph centrality factors:

$$R_{composite} = \sum (w_i \cdot S_i) + \gamma \cdot (\alpha \cdot P_{rank} + \beta \cdot D_{deg}) \quad (1)$$

where S_i represents the normalized probability score generated by modality sub-engine i , w_i denotes the assigned modal weight satisfying $\sum w_i = 1.0$, P_{rank} represents the normalized Neo4j PageRank score of the target entity, D_{deg} represents the normalized node degree centrality, α and β are graph scaling factors, and γ is a global network amplification constant. The resulting raw score is mapped through an isotonic regression calibration layer to produce a final risk percentage $R \in [0, 100\%]$.

Algorithm	1.	Multi-Modal	Sig-
nal	Fusion	& Calibrated	Scoring
Input: Artifact Payload P, Modality M			
Output: Calibrated Risk Score R, SHAP Weights, Narrative N			
1:	P_clean	← PreprocessAndSanitize(P)	
2:	F_lex, F_nlp, F_infra	← ExtractFeatureVectors(P_clean, M)	
3:	S_ml	← XGBoostInference(F_lex, F_infra)	
4:	S_nlp	← TransformerSemanticIntent(F_nlp)	
5:	P_rank, D_deg	← QueryNeo4jCentrality(P_clean)	
6:	R_raw	← $\sum(w_i \cdot S_i) + \gamma \cdot (\alpha \cdot P_{rank} + \beta \cdot D_{deg})$	
7:	R_calibrated	← IsotonicRegressionCalibration(R_raw)	
8:	Φ	← ComputeSHAPAttributions(R_calibrated, F_vectors)	
9:	Context	← QueryChromaDBVectorStore(Φ , P_clean)	
10:	N	← LLMReasoningSynthesis(Context, Φ , R_calibrated)	
11:	Return	{R_calibrated, Φ , N}	

F. Explainable AI (XAI) Framework

To eliminate black-box opacity, TrustNet AI integrates SHapley Additive exPlanations (SHAP). The SHAP methodology calculates additive feature attributions based on game theory, assigning an explicit value ϕ_j to each feature j denoting its positive or negative contribution toward pushing the risk score away from the baseline average:

$$g(z') = \phi_0 + \sum (\phi_j \cdot z'_j) \quad (2)$$

where $g(z')$ represents the explanation model, ϕ_0 is the baseline expectation, and $z'_j \in \{0, 1\}^M$ indicates feature presence. Top positive weights (e.g., "Suspicious subdomain depth (+18%)" or "High urgency phrasing (+24%)") are surfaced visually on the user dashboard.

G. RAG-Based Intelligence Workflow

While SHAP provides quantitative feature weights, non-expert users require plain-English contextual reasoning. TrustNet AI implements a Retrieval-Augmented Generation (RAG) workflow to generate grounded explanations, as illustrated in Fig. 3.



Fig. 3. Retrieval-augmented generation workflow for grounded risk explanation.

The RAG pipeline operates through the following steps:

1) Verified cybersecurity threat advisories, regulatory guidance documents, and historical scam registries are chunked, embedded using vector embedding models, and indexed within a ChromaDB vector database.

2) When a digital signal is analyzed, the top SHAP features and signal metadata form a context query payload.

3) The *Context Retriever* executes cosine similarity search against ChromaDB to extract the top-k relevant threat advisories.

4) The retrieved advisories, original signal payload, and SHAP weights are formatted into a structured prompt passed to the Large Language Model (LLM) reasoning layer.

5) The LLM synthesizes a concise, natural language explanation strictly constrained by the retrieved context, effectively eliminating hallucinations and providing a defensible explanation ("Not just a score — a reason").

H. Result Presentation & Technology Stack

The platform frontend is constructed using React 18, TypeScript, and Tailwind CSS, deployed on Vercel's global edge network. The backend is built on Python 3.10+, FastAPI, Uvicorn, XGBoost, Neo4j Graph DB, and ChromaDB vector store. The user interface renders real-time threat badges, interactive SHAP weight bars, RAG explanation cards, and graph trace visualization panels.

V. NOVELTY AND CONTRIBUTIONS

To establish academic clarity, it is essential to distinguish the novel contributions of this research from the underlying baseline technologies utilized during system implementation. Standard technologies—such as React, FastAPI, Python, XGBoost, Neo4j, and general LLM APIs—represent established software engineering tools rather than scientific contributions in isolation.

The true novelty of TrustNet AI resides in the architectural integration and methodological fusion of these components into a unified threat intelligence framework:

1) *Integrated Multi-Modal Threat Modeling*: Unlike existing security literature that addresses URLs, messages, QR codes, or financial handles in isolation, TrustNet AI introduces an architectural framework capable of cross-correlating signals across five heterogeneous digital modalities within a unified inference engine.

2) *Hybrid Signal Fusion (ML + Graph Centrality)*: The system presents a novel scoring formulation that dynamically combines supervised machine learning probability outputs with graph network topology metrics (Neo4j PageRank and degree centrality), capturing both individual payload anomalies and broader structural syndicate relationships.

3) *Dual-Output Explainable Security Architecture*: TrustNet AI bridges the gap between quantitative model interpretability and qualitative human comprehension by pairing quantitative SHAP feature attribution weights with RAG-grounded natural language narratives.

4) *Factually Anchored RAG Explanation Engine*: By constraining LLM narrative generation to verified vector database

security advisories, the platform prevents AI hallucinations in security-critical decision environments, establishing a defensible paradigm for AI-assisted threat triage.

VI. WEB STRUCTURE AND USER INTERFACE

The user interface of TrustNet AI was deployed to provide a professional, responsive operational console for threat analysis. The application structure enforces a clean, reactive user flow:

User Input Selection → Digital Signal Entry → Asynchronous Analysis Dispatch → Multi-Engine Processing → Calibrated Risk Scoring → RAG Explanation Synthesis → Interactive Evidence Presentation.

The web interface supports navigation tabs corresponding to each digital modality (URL, Message, QR Code, UPI, Phone Number), alongside an integrated Threat Analysis Command Console ("TrustNet OS"). Figure 4 illustrates the actual deployed web interface of TrustNet AI available at the public URL.

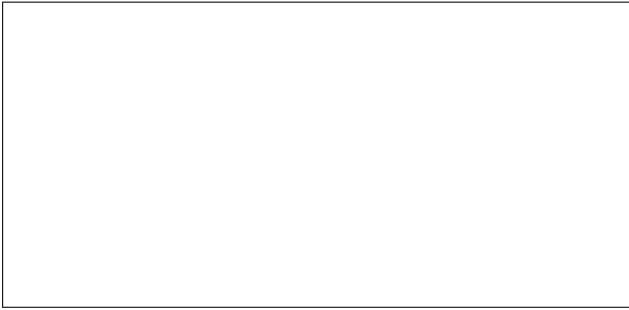


Fig. 4. Web interface of the TrustNet AI threat analysis platform (deployed on Vercel).

As demonstrated in Fig. 4, the user interface integrates a clean, modern aesthetic with high operational functionality. The top header displays system status, telemetry feeds, and navigation controls. The main workspace features an input acquisition panel where users paste target URLs, message texts, or upload QR code images. Upon initiating analysis, the dashboard dynamically populates a risk score gauge, categorical risk level indicator (e.g., High Risk / Critical), SHAP feature weight distribution bars, and the RAG-generated evidence summary card.

VII. RESULTS AND DISCUSSION

In accordance with formal academic reporting standards, this section presents an honest assessment of the current development status and functional evaluation of TrustNet AI. As documented in the Capstone Review baseline, the project is currently approximately 70% complete. Therefore, final empirical validation metrics (such as dataset-wide precision, recall, F1-score, and ROC-AUC curves on full production benchmarks) are currently under active development and are not claimed as finalized results.

The implemented functionality and current platform validation status are categorized as follows:

1) Implemented & Validated Core Functionality (Completed Work):

- Full architectural specification and microservices backend implementation using FastAPI.
- Production deployment of the React + TypeScript frontend web interface on Vercel.
- Operational input acquisition and validation modules across all five digital modalities (URL, Message, QR, UPI, Phone).
- Working feature extraction pipelines for URL lexical analysis, text message n-gram processing, and QR code payload parsing.
- Functional risk scoring interface rendering calibrated numeric scores, risk badges, and SHAP feature attributions.
- Prototype RAG intelligence integration connecting vector database retrieval to LLM narrative generation.

2) Ongoing Development & Model Validation (Current Work):

- *Model Validation & Benchmark Testing*: Conducting systematic cross-validation experiments across large-scale public security benchmarks, including the PhishTank URL dataset, Kaggle SMS Spam Collection, and financial transaction graph datasets.
- *Graph Database Scaling*: Expanding the Neo4j knowledge graph schema to ingest multi-hop transaction topologies and optimize PageRank query latency.
- *RAG Knowledge Base Expansion*: Indexing comprehensive CVE (Common Vulnerabilities and Exposures) databases, CERT security advisories, and anti-phishing registry feeds within ChromaDB.
- *Explainability Refinement & Latency Optimization*: Refining LLM prompt templates to ensure sub-second explanation delivery and minimizing API orchestration overhead.

The planned evaluation methodology will evaluate TrustNet AI across standard performance metrics: Confusion Matrix analysis (True Positives, False Positives, True Negatives, False Negatives), Precision, Recall, F1-Score, ROC-AUC curve area, and RAG hallucination rate evaluations. Initial functional tests demonstrate that the integrated multi-modal pipeline successfully processes input artifacts and renders evidence-backed risk cards across tested sample scenarios.

VIII. CONCLUSION AND FUTURE WORK

This paper presented TrustNet AI, an AI-powered trust and threat intelligence platform designed to address the challenges of multi-channel digital fraud and opaque machine learning security scoring. By establishing a unified intelligence layer capable of processing URLs, natural language text messages, QR codes, UPI handles, and telephone credentials, TrustNet AI breaks down traditional detection silos. The platform combines supervised machine learning, natural language processing, and Neo4j graph centrality metrics (PageRank and degree centrality) to calculate calibrated composite risk scores. Furthermore,

by integrating SHAP feature attributions with a Retrieval-Augmented Generation (RAG) framework, TrustNet AI converts black-box model predictions into transparent, defensible, plain-English security explanations.

The future research and development roadmap for TrustNet AI focuses on completing the remaining 30% of system development prior to final production release:

1) *Rigorous Empirical Validation*: Executing comprehensive benchmark testing on multi-million sample cybersecurity datasets to establish baseline Precision, Recall, and ROC-AUC metrics.

2) *Full RAG & LLM Integration*: Fine-tuning dedicated open-source LLM security models (e.g., Llama-3-Instruct) and expanding the ChromaDB vector database index with real-time threat feed updates.

3) *Advanced Graph Neural Network (GNN) Reasoning*: Upgrading graph analysis from static centrality metrics (PageRank) to dynamic Graph Convolutional Networks (GCNs) for automated fraud ring discovery.

4) *Enterprise SIEM & API Deployment*: Packaging the backend engine into containerized Docker microservices for seamless integration with enterprise Security Information and Event Management (SIEM) systems.

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